作业纸

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若声强级60分贝提升到80分贝

$\Delta L = 10\lg\frac{I_2}{I_1} = 20$

则 $I_2 = 10^2 I_1 = 100 I_1$

即声强提高了100倍

7.20 波源和接收器相对于介质同时且相向运动

$$v_R = \frac{u+v_R}{u-v_S} v_S = \frac{330\,m/s + (-80\,km/h)}{330\,m/s - 120\,km/h} \times 400\,Hz = 415\,Hz$$

7.22 波源静止，接收器靠近波源

$v_S' = \frac{u+v}{u-v} v_S$ 可知 $\Delta v_S = \frac{2v}{u+v} v_S$

$$v = \frac{u\cdot\Delta v_S}{2v_S - \Delta v_S} = 9.47\,m/s$$

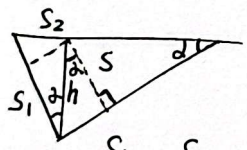
解 $v=$

7.25 (1) 半顶角α

$\sin\alpha = \frac{u}{v} = \frac{1}{2.3}$

$\alpha = \arcsin\frac{1}{2.3} = 25.8°$

(2)



$$\Delta t = \frac{S_1}{u} - \frac{S_2}{v}$$
$$= \frac{S_1}{u} - \frac{S_2}{u}\sin\alpha$$
$$= \frac{S}{u}$$
$$= \frac{h\cos\alpha}{u}$$
$$= \frac{5000\,m \times \sqrt{1-\left(\frac{1}{2.3}\right)^2}}{330}$$
$$= 13.64\,s$$

7.29 $w^2 = w_p^2 + c^2k^2$

$$u = \frac{w}{k} = \frac{\sqrt{w_p^2+c^2k^2}}{k}$$

$$u_g = \frac{dw}{dk} = \frac{c^2k}{\sqrt{w_p^2+c^2k^2}}$$

$uu_g = c^2$ 易证

9.1 (1) $P = \frac{373.15\,K}{273.16\,K} \times 6.65\times10^3\,Pa$

$= 9.08\times10^3\,Pa$

(2) $T = \frac{2.20\times10^3\,Pa}{6.65\times10^3\,Pa} \times 273.16\,K$

$= 90.4\,K$

$= -182.8\,°C$

9.3 $V_1 = \pi \times 71.12\,cm \times \pi \times 1.5\,cm^2$

$V_2 = 30\,cm \times \pi \times 1.5^2\,cm^2 \times 20$

$\frac{P_0V_2}{T_0} = \frac{P_1V_1}{T_1}$　　$T_0 = 270\,K$　$T_1 = 280\,K$

解 $P_1 = 2.8P_0$

$= 2.8\,(atm)$

9.8 (1) $PV = nRT = \frac{m}{M}RT$ 可得 $PM = \rho RT$

$(p+dp)S + \rho gS\cdot dh = pS$

即 $dp = -\rho g dh$

$dp = -\frac{PMg}{RT}dh \Rightarrow \frac{dp}{p} = -\frac{Mg}{R(T_0-\alpha h)}dh$

两边积分有

$\ln\frac{p}{p_0} = +\frac{Mg}{R\alpha}[\ln(T_0-\alpha h) - \ln T_0]$

$p = p_0 e^{\frac{Mg}{R\alpha}[\ln(T_0-\alpha h)-\ln T_0]}$

$= p_0 \exp\left[\frac{Mg}{R\alpha}\ln\left(1-\frac{\alpha h}{T_0}\right)\right]$

(此处 h 即为 y)

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(2) $\alpha \to 0$时

$$\lim_{\alpha\to 0} P = \lim_{\alpha\to 0} P_0 \exp\left[\frac{Mg}{R}\,\frac{\ln(1-\frac{\alpha y}{T_0})}{\alpha}\right]$$

$$= \lim_{\alpha\to 0} P_0 \exp\left(\frac{Mg}{R}\cdot\frac{-\frac{y}{T_0}\alpha}{\alpha}\right)$$

$$= P_0\, e^{-\frac{Mgy}{RT}}$$

(此处 y 即 9.16 中的 h)

(3) $y = 8844.43\,m$

$T = T_0 - \alpha y = (273 - 0.6\times10^{-2}\times 8844.43)K = -53\,°C$

$= 220\,K$

$$P = P_0\, e^{-\frac{Mgy}{RT}} = 1\times e^{-\frac{29.0\times9.8\times8844.43}{8.31\times220}}\ atm$$

$= 0.29\ (atm)$

9.10 $\overline{(v-\bar{v})^2} = \overline{v^2+\bar{v}^2-2v\bar{v}} = \overline{v^2}+\bar{v}^2-2\bar{v}^2 = \overline{v^2}-\bar{v}^2 \geq 0$

则 $\overline{v^2} \geq \bar{v}^2$

即 $\sqrt{\overline{v^2}} \geq \bar{v}$

9.14 (1) $\sigma = \pi\left(\frac{d_{电}+d}{2}\right)^2$

又 $d_{电} \ll d$

则 $\sigma = \frac{\pi d^2}{4}$

(2) $\bar{z} = n\sigma\bar{u}$

$\bar{u} = \bar{v}_e$ (气体分子可以认为是不动的)

$\bar{\lambda} = \frac{\bar{v}_e}{\bar{z}}$

解 $\bar{\lambda} = \frac{1}{n\sigma} = \frac{4}{\pi d^2 n}$

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